Erez A. Zimmerman

Ph.D. candidate in Astrophysics, Weizmann Institute of Science
Supernovae | Time Domain Astronomy | Circumstellar Matter

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Research Profile

I study supernovae to constrain their progenitor stars, pre-explosion mass loss, and explosion mechanisms. In particular, I have studied nearby supernovae, including the decadal supernova **2023ixf** (Zimmerman et al. 2024, published in Nature), for which I obtained first-of-its-kind ultraviolet spectroscopy using the *Hubble Space Telescope*, and the extremely faint Iax supernova **2024vjm** (Zimmerman et al., accepted in principle at Nature Astronomy), for which I analyzed a pre-explosion *Euclid* space telescope image containing the explosion site. I have led several successful telescope proposals for facilities such as the **Hubble Space Telescope**, **Gemini**, and the **VLT**.

Research Leadership and Collaborations

- I am a member of major transient collaborations including **BlackGEM**, **LAST**, **ePESSTO+**, **SoXS**, **ULTRASAT** and formerly **ZTF**.
- I lead the BlackGEM consortium bi-weekly transient meetings
- I developed **CAST**, a TOM-based transient marshal for the Weizmann Observatory's LAST collaboration.
- I am a member of the ULTRASAT Marshal development team.

Education

Weizmann Institute of Science — Rehovot, Israel
Ph.D. in Astrophysics; advisor: Prof. Avishay Gal-Yam — 2021 – expected Oct. 2026

Weizmann Institute of Science — Rehovot, Israel
M.Sc. in Astrophysics; advisor: Prof. Avishay Gal-Yam — 2018 – 2021

Technion – Israel Institute of Technology — Haifa, Israel
B.Sc. in Physics — 2015 – 2018

Honors and Awards

Weizmann School of Science Prof. Israel Dostrovsky Prize of Excellence — 2026
Weizmann School of Science Haim Holtzman Memorial Prize — 2025

First and Second Author Publications

The Complex Circumstellar Environment of Supernova 2023ixf
Erez A. Zimmerman, Ido Irani, Ping Chen, Avishay Gal-Yam et al.
Nature, 627, 759–762, 2024

A Faint Progenitor System for the Faint Supernova 2024vjm
Erez A. Zimmerman, Avishay Gal-Yam et al.
Nature Astronomy, accepted in principle

The Type IIn SN 2025cbj coincidence with the high-energy neutrino IceCube-250421A
Simone Garrappa, **Erez A. Zimmerman**, Tal Wasserman et al.
A&A, 708, A223 (2026)

Co-Author Publications

XSNAP: An X-ray Supernova Analysis Pipeline with Application to the Type II Supernova 2024ggi
Ferdinand, Wynn V. Jacobson-Galán, Mansi M. Kasliwal, **Erez A. Zimmerman**
ApJ, 1001, 26 (2026)

The flash-ionised SN Ibn 2025kzr: H-free circumstellar material formed during a precursor outburst 55 days prior to explosion
S. de Wet, G. Leloudas, D. A. H. Buckley, N. Erasmus, P. J. Groot, **Erez A. Zimmerman**, et al.
A&A, accepted (2026)

SN 2019vxm: A Shocking Coincidence between Fermi and TESS
Zachary G. Lane et al. including **Erez A. Zimmerman**
ApJ, 1003, 19 (2026)

The Large Array Survey Telescope Pipeline: II. Image subtraction and transient detection
R. Konno et al. including **Erez A. Zimmerman**
A&A, 710, A406 (2026)

Extremely Stripped Supernova Reveals a Silicon and Sulfur Formation Site
Steve Schulze et al. including **Erez A. Zimmerman**
Nature, 644, 634–639 (2025)

A low mass, binary-stripped envelope for the Type IIb SN 2024abfo
S. de Wet, G. Leloudas, D. A. H. Buckley, N. Erasmus, P. J. Groot, **Erez A. Zimmerman**
A&A, 704, A89 (2025)

A Search for Minute-timescale Flares from the Transient AT 2024wpp
Eran O. Ofek et al. including **Erez A. Zimmerman**
ApJ, 993, 76 (2025)

Low-luminosity Type IIP Supernovae from the Zwicky Transient Facility Census of the Local Universe. I. Luminosity Function, Volumetric Rate
Kaustav K. Das et al. including **Erez A. Zimmerman**
PASP, 137, 044203 (2025)

Spectroscopic observations of progenitor activity 100 days before a Type Ibn supernova
S. J. Brennan, J. Sollerman, I. Irani, S. Schulze, P. Chen, et al. including **E. A. Zimmerman**
A&A, 684, L18 (2024)

The Progenitor Star of SN 2023ixf: A Massive Red Supergiant with Enhanced, Episodic Pre-Supernova Mass Loss
Yu-Jing Qin, Keming Zhang, Joshua Bloom, Jesper Sollerman, **Erez A. Zimmerman**, et al.
MNRAS, 534, Issue 1, 271–280 (2024)

The BlackGEM Telescope Array. I. Overview
Paul J. Groot et al. including **Erez A. Zimmerman**
PASP, 136, 115003 (2024)

The Early Ultraviolet Light Curves of Type II Supernovae and the Radii of Their Progenitor Stars
Ido Irani et al. including **Erez A. Zimmerman**
ApJ, 970, 96 (2024)

SN 2022oqm–A Ca-rich Explosion of a Compact Progenitor Embedded in C/O Circumstellar Material
Ido Irani et al. including **Erez A. Zimmerman**
ApJ, 962, 109 (2024)

The Prevalence and Influence of Circumstellar Material Around Hydrogen-rich Supernova Progenitors
Rachel J. Bruch, Avishay Gal-Yam, Ofer Yaron, Ping Chen, Nora L. Strotjohann, Ido Irani, **Erez Zimmerman**, et al.
ApJ, 952, 119 (2023)

A WC/WO Star Exploding Within an Expanding Carbon–Oxygen–Neon Nebula
Avishay Gal-Yam et al. including **Erez A. Zimmerman**
Nature, 601, 201–204 (2022)

Candidate Tidal Disruption Event AT2019fdr Coincident with a High-Energy Neutrino
Simeon Reusch et al. including **Erez Zimmerman**
Phys. Rev. Lett. 128, 221101 (2022)

The Spectacular Ultraviolet Flash from the Peculiar Type Ia Supernova 2019yvq
A. A. Miller, M. R. Magee, A. Polin, K. Maguire, **E. Zimmerman** et al.
ApJ, 898, 56 (2020)

Accepted Telescope Time as PI

Infant Supernovae: Observing Massive Stars as They Explode with the Large Array Survey Telescope
Gemini 2025B
4.5 hours

Observing the Local Environment of the Nearby Faintest Iax SN 2024vjm with MUSE Narrow Field Mode
ESO Period 114 DDT
2.9 hours, VLT/MUSE

Observing the Circumstellar Interaction of the 7 Mpc SN 2024ggi Chandra DDT 25508981
30 ks

High-Resolution Spectroscopy of One of the Nearest Supernovae in a Decade: SN 2024ggi ESO Period 113 DDT
2.7 hours, VLT/UVES

Explosions in Real Time: Rapid UV Supernova Flash Spectroscopy HST Cycle 30
22 orbits

Infant Supernovae: Observing Massive Stars as They Explode OPTICON 2024B
Liverpool Telescope 5 hr + TNG 0.3 nights

Infant Supernovae: Observing Massive Stars as They Explode from the Ground and Space Gemini 2023A
0.9 nights

Explosions in Real Time: Rapid UV Supernova Flash Spectroscopy OPTICON 2023A
TNG 1.0 night

Observational Experience

- **Hubble Space Telescope**: 22 orbits.
- **VLT/MUSE**: 1 remote observing run.
- **VLT/UVES**: 4 remote observing runs.
- **NTT, La Silla Observatory, Chile**: 9.5 nights + ToO observations in ePessto+
- **Lowell Discovery Telescope, Lowell Observatory, Arizona, USA**: 1 night as well as ToO observations.
- **Neil Gehrels Swift Observatory**: 40+ approved ToO observations
- **Aristarchos Telescope, Chelmos Observatory, Greece**: 6 nights.
- **Target-of-opportunity activations**: extensive experience triggering rapid ToO photometry and spectroscopy with multiple facilities and instruments.

Teaching

Lecturer, Open University of Israel 2022 – 2026
Secrets of the Universe: A Window to Astronomy

Instructor for a general astronomy course covering the history of astronomy, the Solar System, astronomical instrumentation, stellar physics, supernovae and supernova remnants, galaxy formation, and cosmology. Average student rating: **4.3/5**.

Invited Talks

SN 2023ixf Early Emission and Circumstellar-Material Mapping Jun. 2024
Review talk, SN 2023ixf: The Decadal Supernova in M101 Workshop, European Southern Observatory, Garching, Germany.

Ultraviolet Observations Resolve the Explosion of Supernova 2023ixf in Messier 101 within Its Complex Circumstellar Environment Jan. 2024
American Astronomical Society Meeting 243, New Orleans, LA, USA.

Invited Seminars

Dying Stars Shed Light on Late-Stage Mass Loss Nov. 2025
Invited seminar, Technion, Haifa, Israel.

A Framework to Map Circumstellar Material Around Core-Collapse Supernovae May 2025
Invited seminar, Hebrew University of Jerusalem, Jerusalem, Israel.

The Complex Circumstellar Environment of Supernova 2023ixf Feb. 2024
Tel Aviv University Student Seminar, Tel Aviv University, Tel Aviv, Israel.

Selected Contributed Talks

Dying Stars Shed Light on Late-Stage Mass Loss Oct. 2026
Type II Supernovae: The Chaotic Deaths of Supergiants and their Cosmic Implications, San Pedro de Atacama, Chile.

A Faint Progenitor System for the Faint Supernova 2024vjm Jun. 2026
American Astronomical Society Meeting 248, Pasadena, CA, USA.

Dying Stars Shed Light on Late-Stage Mass Loss Nov. 2025
Stellar Populations and Their Explosions: Bridging the Gap Workshop, Max Planck Institute for Astrophysics, Ringberg Castle, Germany.

SN 2024vjm: A Faint Supernova and a Fainter Companion Sep. 2025
BlackGEM Consortium Fall Meeting 2025, Hamburg Observatory, Hamburg, Germany.

A Framework to Map the Circumstellar Material Around Red Supergiant Supernovae Jul. 2025
Israel Physical Society Meeting, Technion, Haifa, Israel.

Ultraviolet Observations of Infant Supernovae Mar. 2025
Transients from Space, STScI, Baltimore, MD, USA.

SN 2024ggi and Infant Supernovae: Early Light Curve and Circumstellar Material Sep. 2024
BlackGEM Consortium Fall Meeting 2024, University of Manchester, Manchester, UK.

Early and UV Observations of SN 2023ixf, a Core-Collapse Supernova in the Nearby M101 Galaxy Aug. 2023
SNeX – SuperNova EXplosions Conference, Technion, Haifa, Israel.

High-Velocity Circumstellar Wind Around the Type II Supernova SN 2020pni Jun. 2022
Zwicky Transient Facility Spring Meeting 2022, CNRS/IN2P3, Paris, France.

Posters

A Faint Progenitor System for the Faint Supernova 2024vjm May 2026
IAU Symposium 406 – Future Landscape of Astrophysical Transients, Turku, Finland.

The Complex Circumstellar Environment of Supernova 2023ixf Aug. 2024
International Astronomical Union General Assembly, Cape Town, South Africa, remote.

High-Velocity Circumstellar Wind Around the Type II Supernova SN 2020pni Feb. 2022
Israel Physical Society 67th Meeting, Ben-Gurion University, Be'er Sheva, Israel.

Outreach, Service, and Media

- I peer-review manuscripts for the Astrophysical Journal (ApJ) and Astronomy & Astrophysics (A&A).
- Weizmann student mission to Latin America in Chile and Brazil — August 2026
- Recorded lecture for Israeli Space Week, *Telescopes in Space Reveal the Mysteries of the Universe*, for ages 13–14, Hebrew — Jan. 2025
- Guest lecture for the Israeli Astronomical Society, *We Are All Stardust: Supernovae and the Origin of the Elements*, Hebrew — Sep. 2024
- Guest, *Voice of the Universe* podcast, episode 1, Hebrew — Apr. 2024
- Guest, BBC *Science in Action*, "Brightest Supernova in a Decade" — May 2023
- Press coverage of Zimmerman et al. 2024 in Weizmann, Stockholm University, Keck Observatory, Ynet, Mako, and Kan Radio.

Other Professional Experience and Affiliations

Weizmann–Helmholtz Multi-Messenger School Fellow	2021 – 2026
Summer Student at the European Organization for Nuclear Research (CERN)	Jun. 2018 – Sep. 2018
Secretary general of the Technion student union	2017
Head of physics student council, Technion	2016-2017
Mandatory Military Service	2012-2015